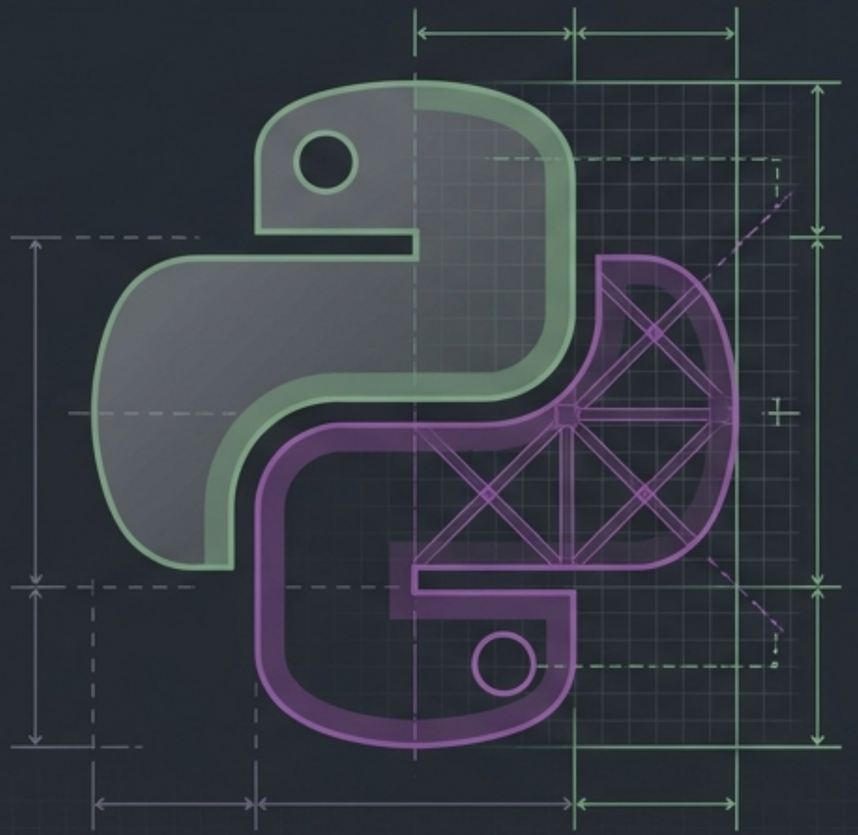
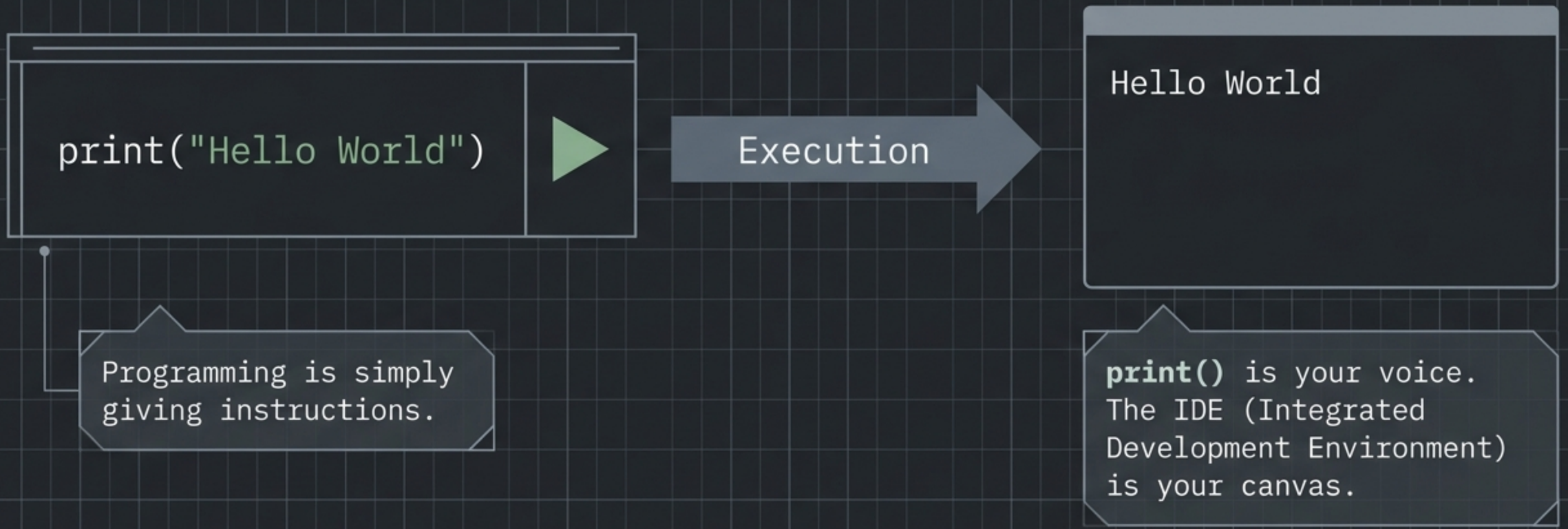
The Python Blueprint

From First Script to First AI: A 30-Minute Architecture Course



Gemini Notebook

Speaking the Machine's Language



Gemini Notebook

Variables: Naming Your Memory



Variables store data so you can reuse and update it dynamically.

```
print("Hello " + name + "!")
```



Rigid, older method prone to errors.

```
print(f"Hello {name}!")
```

F-Strings (`f"..."`) allow you to seamlessly drop variables directly into your text using curly brackets {}.

Gemini Notebook

The Raw Materials (Atomic Data Types)

Python needs to know the exact material you are working with. You cannot do math with a String, just as you wouldn't build a foundational wall out of glass.

String (Text)



"Apple"
"Bob"

Integer (Whole Numbers)



10
-10

Float (Decimals)



10.5, 3.1415

Boolean (Binary State)

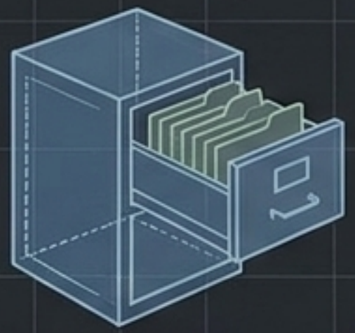


True, False

Gemini Notebook

Structuring Complex Data

List `[]`



Mutable, Ordered

```
names =
  ['Agnetha',
  'Björn',
  'Benny',
  'Anni-Frid']
```

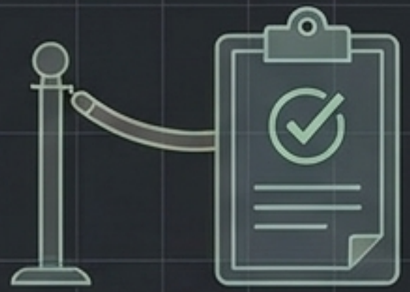
Tuple `()`



Immutable, Ordered

```
coordinates =
  (2.5, 1.5)
```

Set `{}`



Mutable, No Duplicates

```
numbers = {1, 2, 3,
  3, 4, 4, 5}

print(numbers)
# Output: {1, 2, 3, 4, 5}
```

Dictionary {k:v}`

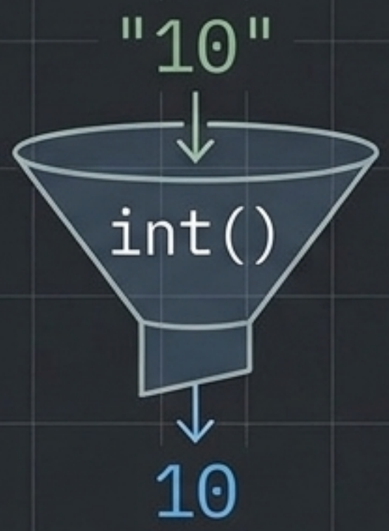


Key-Value pairs

```
scores =
  {'Bob': 1,
  'James': 2}
```

Gemini Notebook

Enforcing the Rules (Type Safety)



Converting messy scraped text into usable math.



Annotations don’t change how code runs, but they act as digital guardrails, warning the developer if they try to put text inside a number variable.

Gemini Notebook

The Logic Gates: Making Comparisons

Before a program can make a decision, it must weigh the conditions. Every complex comparison ultimately reduces to a single binary state: **True** or **False**.

`1 > 2`



→ `False`

`1 < 2`



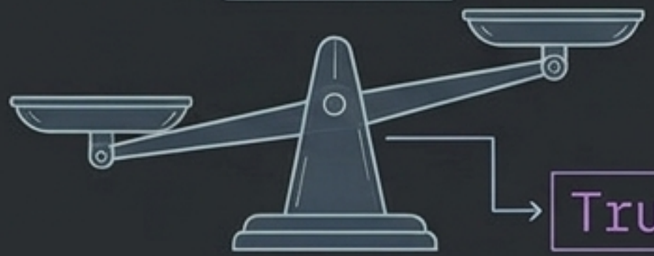
→ `True`

`2 ≥ 2`



→ `True`

`1 ≤ 2`



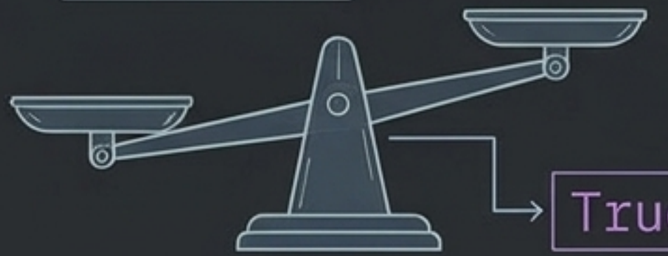
→ `True`

`1 == 1` (Is exactly equal to)



→ `True`

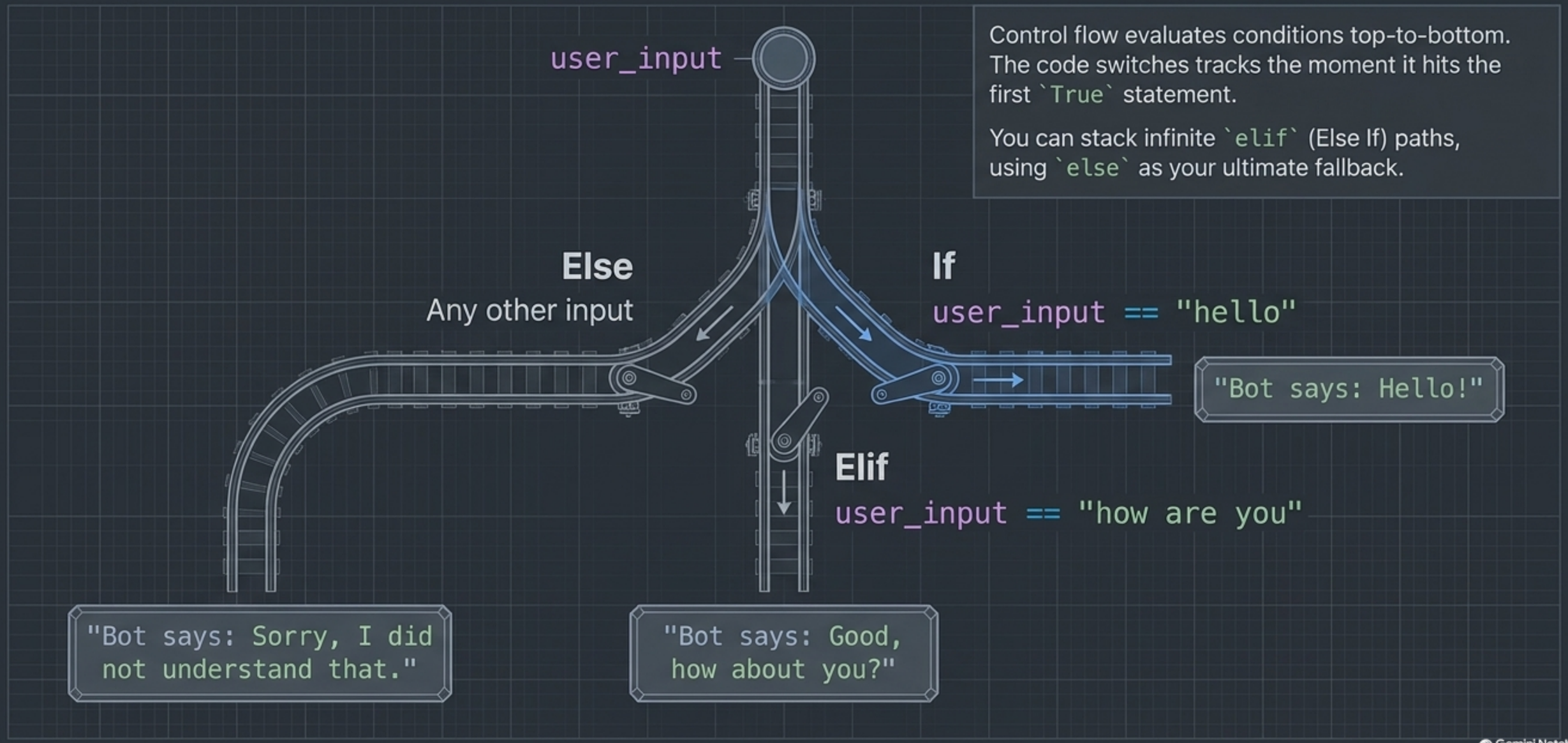
`1 != 2` (Is NOT equal to)



→ `True`

Gemini Notebook

Forking the Path (Control Flow)



Control flow evaluates conditions top-to-bottom. The code switches tracks the moment it hits the first `True` statement.

You can stack infinite `elif` (Else If) paths, using `else` as your ultimate fallback.

Gemini Notebook

The Engines of Automation: For vs. While

For Loop




```
for i in range(3):
    print("hello")
```

Finite iteration: Runs exactly as many times as the items in a sequence (or list).

While Loop

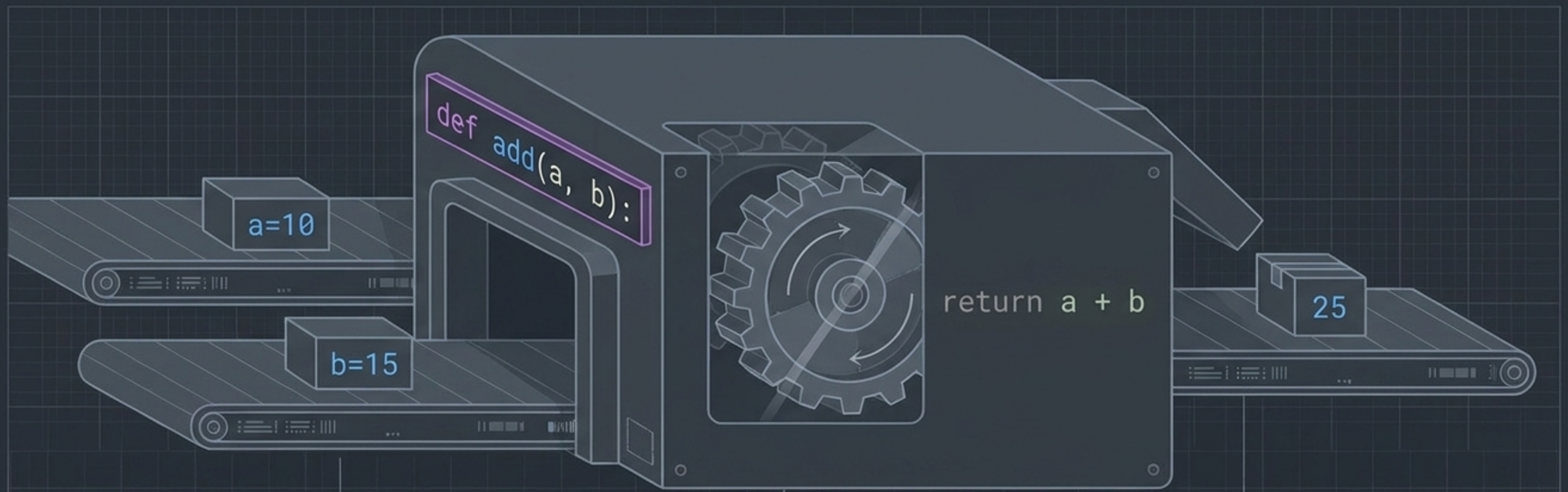


```
while True:
    print("hello")
```

Infinite iteration: Runs forever until a specific condition forces the loop to break or evaluate to ‘False’.

Gemini Notebook

Functions: Reusable Logic Factories



Stop repeating yourself. Functions encapsulate logic into a reusable machine.

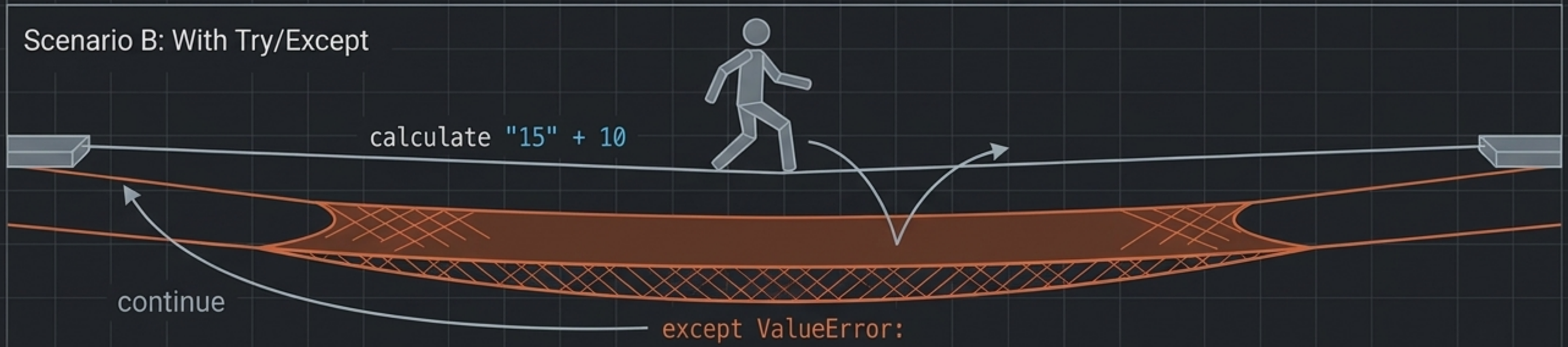
You can set default behaviors directly on the intake (e.g., `def greet(name, greeting='Hi')`). If a greeting isn't provided, the factory uses its default.

Gemini Notebook

The Safety Net (Try / Except)

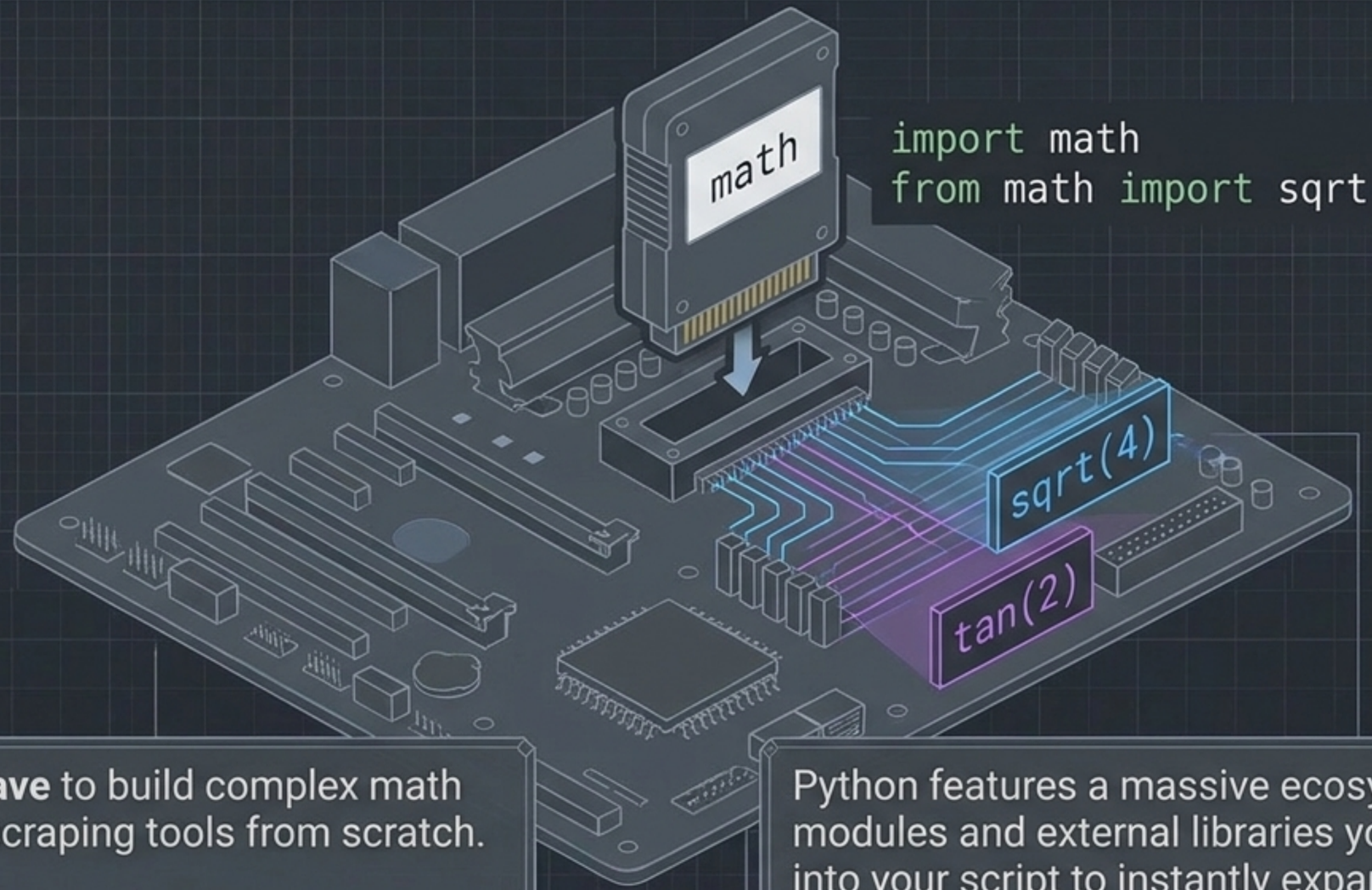
Users will break your app. Professional developers anticipate this. Wrapping dangerous code in a `try` block allows you to catch specific errors and ask the user to fix their input, rather than letting the entire application fatally crash.





Gemini Notebook

Borrowing Superpowers (External Imports)

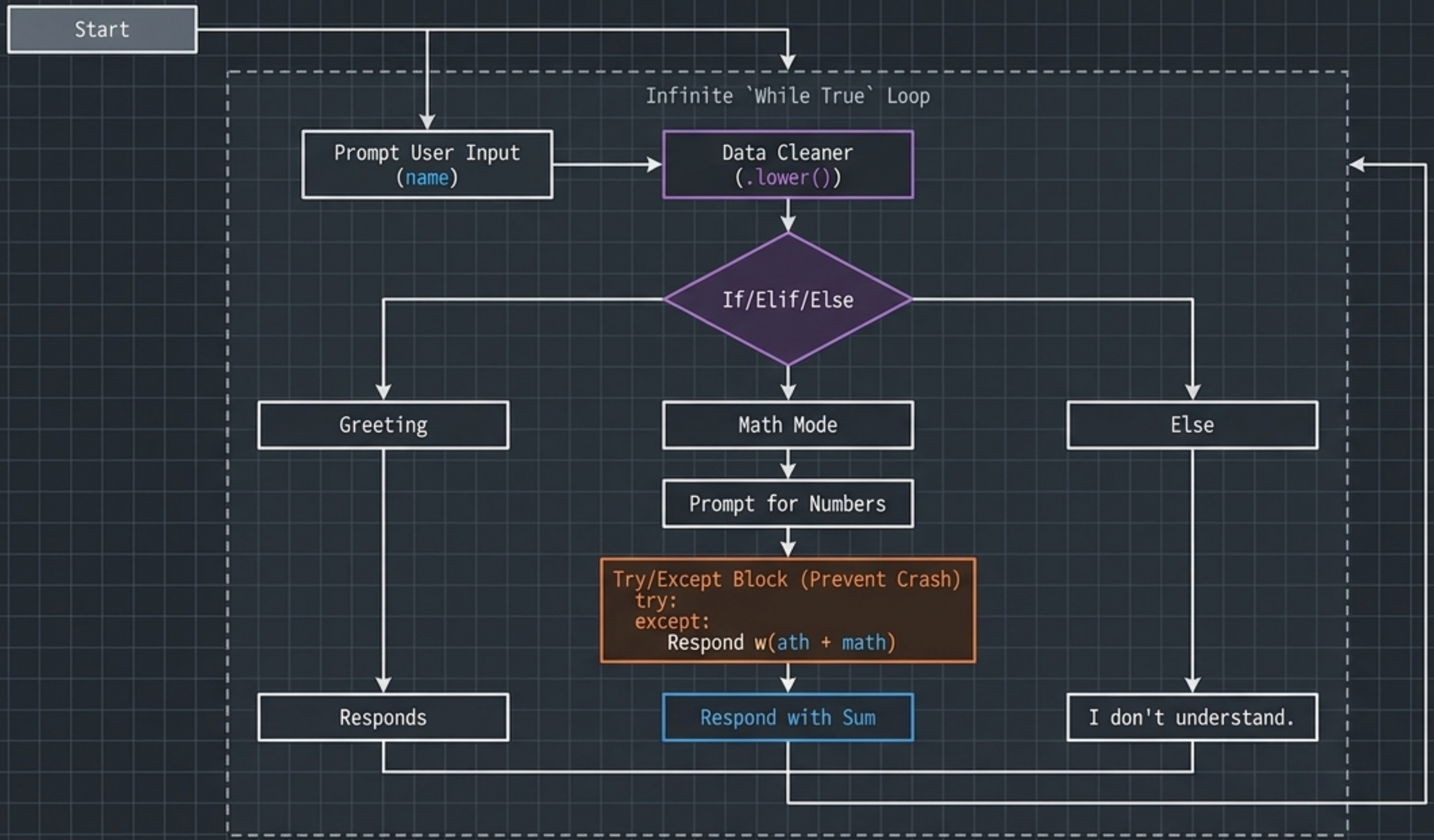


You don't have to build complex math or internet-scraping tools from scratch.

Python features a massive ecosystem of built-in modules and external libraries you can plug directly into your script to instantly expand its capabilities.

Gemini Notebook

Synthesis Phase I: Chatbot Architecture



Gemini Notebook

Synthesis Phase II: The Finished Machine

1



The infinite loop engine keeping the bot alive.

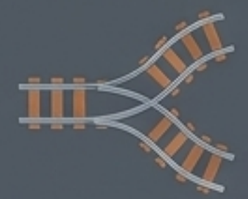
2



Standardizing data so 'Hi' and 'hi' are treated equally.

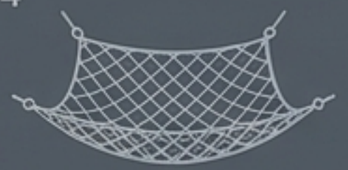
```
while True:
    user_input = input("You: ").lower()
    if user_input in ['hi', 'hello']:
        print("Bot: Hi there!")
    elif user_input in ['add', '+']:
        try:
            num1 = float(input("First number: "))
            num2 = float(input("Second number: "))
            print(f"Bot: {num1 + num2}")
        except ValueError:
            print("Bot: Invalid number.")
    else:
        print("Bot: I don't understand.")
```

3



Control flow switching to math mode.

4

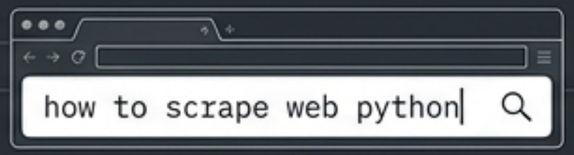


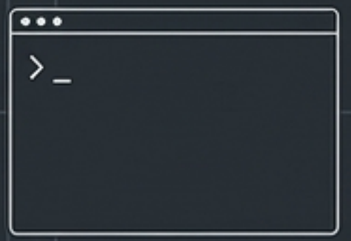
Catching letters when expecting numbers to prevent a crash.

Gemini Notebook

The Developer’s Mindset







1 Start Small

Don’t try to build Facebook on day one. Master the basics first.

2 Google Everything

Even developers with 5 years of experience Google syntax constantly. It is part of the job.

3 Build Real Apps

The only way to truly learn is to pick a project you care about, encounter errors, and solve them.

You now have the fundamental blueprint. Welcome to Python.

Gemini Notebook